Mathematical Methods MTH1021 Problem sheet 22

Volume and surface integrals. Gauss's and Stokes's theorems.

[SUBMIT ANSWERS TO ALL QUESTIONS]

Volume integral: $\int\limits_V f dV$ or $\iiint\limits_V f(x,y,z)dxdydz$ (in Cartesian coordinates).

Surface integral of the first kind $\int\limits_S f dS$; surface integral of the second kind $\int\limits_S \boldsymbol{A}\cdot d\boldsymbol{S}$, where $d\boldsymbol{S} = \boldsymbol{n}dS$ and $\boldsymbol{n}$ is a unit vector normal to the surface.

For a surface in parametric form, $\boldsymbol{r} = \boldsymbol{r}(u,v)$, the surface element is $d\boldsymbol{S} = \frac{\partial \boldsymbol{r}}{\partial u} \times \frac{\partial \boldsymbol{r}}{\partial v} du dv$.

Gauss's theorem:

$$\oint\limits_S \boldsymbol{A}\cdot d\boldsymbol{S} = \int\limits_V \nabla\cdot\boldsymbol{A}dV,$$

where V is the volume enclosed by the surface S, and $d\boldsymbol{S}$ is outwards from V.

Stokes's theorem:

$$\oint\limits_C \boldsymbol{A}\cdot d\boldsymbol{r} = \int\limits_S \nabla\times\boldsymbol{A}\cdot d\boldsymbol{S},$$

where S is a surface that spans the closed curve C, and where the direction of $d\boldsymbol{S}$ is related to the direction in which C is traversed by the right-handed corkscrew rule.

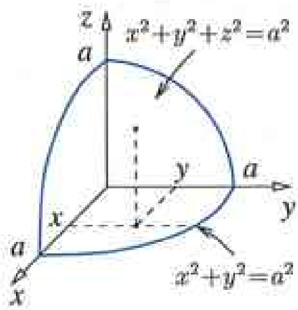
---

1. (a) Evaluate $\iiint\limits_V xyz dxdydz$, where V is defined by $0 \le x \le 1$, $0 \le y \le x$, $0 \le z \le xy$. [6]

   (b) Evaluate the integral $\int_0^1 dx \int_0^1 dy \int_0^1 \frac{dz}{\sqrt{x+y+z+1}}$. [7]

   (c) Evaluate $\iiint\limits_V e^{x+y+z}dxdydz$, where V is a tetrahedron bounded by the planes $x+y+z=1$, $x=0$, $y=0$, $z=0$. [7]

2. Evaluate the volume integral $\int\limits_V xyz dV$, where V is bounded by the surfaces $x^2+y^2+z^2=a^2$ (a sphere of radius a), $x=0$ (the $y-z$ plane), $y=0$ (the $x-z$ plane), $z=0$ (the $x-y$ plane), and located in the first octant, i.e., for $x\ge 0$, $y \ge 0$ and $z\ge 0$, (see diagram)
   (a) using Cartesian coordinates;
   (b) using spherical coordinates. [10]
   [10]



3. Consider a vector field $\boldsymbol{A} = x^2yz\boldsymbol{i} + xy^2z\boldsymbol{j} + xyz^2\boldsymbol{k}$.

   (a) Evaluate $\oint\limits_S \boldsymbol{A}\cdot d\boldsymbol{S}$ for the surface S that bounds volume V in Question 2. [10]

   (b) Find $\nabla\cdot\boldsymbol{A}$ and obtain $\oint\limits_S \boldsymbol{A}\cdot d\boldsymbol{S}$ from Gauss's theorem using the result of Question 2. [10]

4. (a) For the vector field $\boldsymbol{B} = y^2\boldsymbol{i} + z^2\boldsymbol{j} + x^2\boldsymbol{k}$, evaluate $\int_S \boldsymbol{B} \cdot d\boldsymbol{S}$, where S is

i. a hemisphere, $x^2 + y^2 + z^2 = a^2$, $z \geq 0$, with $d\boldsymbol{S}$ along $\boldsymbol{r}$;

[10]

ii. a disk, $x^2 + y^2 \leq a^2$, $z = 0$, with $d\boldsymbol{S}$ along $\boldsymbol{k}$.

[10]

(b) For $\boldsymbol{A} = \frac{1}{3}(z^3\boldsymbol{i} + x^3\boldsymbol{j} + y^3\boldsymbol{k})$, find $\oint_C \boldsymbol{A} \cdot d\boldsymbol{r}$ for the curve C: $x^2 + y^2 = a^2$, $z = 0$, traversed anticlockwise when viewed from positive z.

[10]

(c) Show that $\boldsymbol{B} = \nabla \times \boldsymbol{A}$, and use Stokes's theorem to explain the relation between the answers in parts (a) and (b).

[10]

Problem Sheet 22 MTH1021

Volume and Surface Integrals: Gauss's and Stoke's theorem.

Solutions.

Q1

(a) $\iiint_V xyz\,dx\,dy\,dz$ where V is defined by $0 \le x \le 1$, $0 \le y \le x$, $0 \le z \le xy$.

This volume integral can be written as

$$\int_0^1 dx \int_0^x dy \int_0^{xy} xyz\,dz$$

$$= \int_0^1 x\,dx \int_0^x y\,dy \int_0^{xy} z\,dz$$

The integral on the right with respect to z must be considered first, then over y, last over x.

$$= \int_0^1 x\,dx \int_0^x y\,dy \left[\frac{z^2}{2}\right]_0^{xy}$$

$$= \frac{1}{2}\int_0^1 x\,dx \int_0^x y\,dy\,(x^2y^2 - 0)$$

$$= \frac{1}{2}\int_0^1 x^3\,dx \int_0^x y^3\,dy$$

$$= \frac{1}{2}\int_0^1 x^3\,dx \left[y^4/4\right]_0^x$$

$$= \frac{1}{8}\int_0^1 x^3\,dx\,\left[x^4\right] = \frac{1}{8}\int_0^1 x^7\,dx$$

$$= \frac{1}{8}\left[\frac{x^8}{8}\right]_0^1 = \frac{1}{8}\left(\frac{1}{8} - 0\right)$$

$$= \frac{1}{64}.$$

(b). $\int_0^1 dx \int_0^1 dy \int_0^1 \frac{dz}{\sqrt{x+y+z+1}}$

When integrating over z, x and y are considered as constants. Hence if

$$u = x+y+z+1$$

$$\frac{du}{dz} = 1 \Rightarrow du = dz$$

and the integral becomes

$$\int \frac{du}{\sqrt{u}} = 2\sqrt{u} + C.$$

$$= \int_0^1 dx \int_0^1 dy \left[2\sqrt{x+y+z+1}\right]_0^1$$

$$= 2\int_0^1 dx \int_0^1 dy \left(\sqrt{x+y+2} - \sqrt{x+y+1}\right)$$

Here when integrating over y, we regard x as constant. Hence

$$u = x+y+2 \quad \text{or} \quad u = x+y+1$$

$$du = dy \qquad du = dy$$

and the integrals become

$$\int u^{1/2}\,du \quad \text{and} \int u^{1/2}\,du.$$

$$= \frac{2u^{3/2}}{3}, \quad \frac{2u^{3/2}}{3}.$$

$$= 2\int_0^1 dx \left[\frac{2}{3}(x+y+2)^{3/2} - \frac{2}{3}(x+y+1)^{3/2}\right]_0^1$$

$$= 2\int_0^1 dx\, \frac{2}{3}\left[(x+3)^{3/2} - (x+2)^{3/2} - (x+2)^{3/2} + (x+1)^{3/2}\right]$$

$$= \frac{4}{3}\int_0^1 \left[(x+3)^{3/2} - 2(x+2)^{3/2} + (x+1)^{3/2}\right] dx.$$

Now $\int u^{3/2}\,du = \frac{2}{5}u^{5/2} + C.$

$$= \frac{4}{3}\cdot\frac{2}{5}\left[(x+3)^{5/2} - 2(x+2)^{5/2} + (x+1)^{5/2}\right]_0^1$$

$$= \frac{8}{15}\left[4^{5/2} - 2.3^{5/2} + 2^{5/2} - 3^{5/2} + 2.2^{5/2} - 1^{5/2}\right]$$

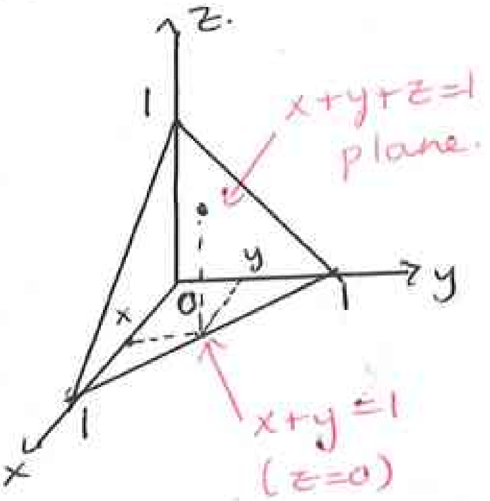
$$= \frac{8}{15}\left[32 - 27\sqrt{3} + 12\sqrt{2} - 1\right]$$

$$= \frac{8}{15}\left(31 - 27\sqrt{3} + 12\sqrt{2}\right)$$

(c). $\iiint_V e^{x+y+z}\, dx\, dy\, dz$

V is tetrahedron bounded by planes $x+y+z=1$, $x=0$, $y=0$, $z=0$.

The plane $x+y+z=1$ cuts the x, y, and z axes at $x=1$, $y=1$, $z=1$ and is plotted below.



The range of integration is

$0 \le x \le 1$

$0 \le y \le 1-x$ (for a given x)

$0 \le z \le 1-x-y$.

(for a given x and y).

Hence we have

$$\iiint_V e^{x+y+z}\, dx\, dy\, dz = \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} e^{x+y+z}\, dz$$

$$= \int_0^1 dx \int_0^{1-x} dy \left[e^{x+y+z}\right]_0^{1-x-y}$$

$$= \int_0^1 dx \int_0^{1-x} dy \left[e - e^{x+y}\right]$$

$$= \int_0^1 dx \left[ey - e^{x+y}\right]_0^{1-x}$$

$$= \int_0^1 dx \left[e(1-x) - e - 0 + e^x \right]$$

$$= \int_0^1 dx \left[e^x - e x \right] = \left[e^x - e\frac{x^2}{2} \right]_0^1$$

$$= e - e \cdot \tfrac{1}{2} - 1 + 0 = \frac{e}{2} - 1.$$

Q2. $\int_V xyz \, dV.$

(a)

The volume V in this question is part of a sphere of radius a, bounded by the spherical surface $x^2+y^2+z^2 = a^2$, and located in the first octant $(x \geq 0, y \geq 0, z \geq 0)$.

For a given x, $0 \leq x \leq a$

y can take values $0 \leq y \leq \sqrt{a^2-x^2}$

and for a given pair of values (x, y), z can take values $0 \leq z \leq \sqrt{a^2-x^2-y^2}$.

Hence in Cartesian co-ordinates the integral is written as

$$\int_V xyz \, dV = \int_0^a dx \int_0^{\sqrt{a^2-x^2}} dy \int_0^{\sqrt{a^2-x^2-y^2}} xyz \, dz$$

$$= \int_0^a x \, dx \int_0^{\sqrt{a^2-x^2}} y \, dy \left[\frac{z^2}{2} \right]_0^{\sqrt{a^2-x^2-y^2}}$$

$$= \frac{1}{2} \int_0^a x \, dx \int_0^{\sqrt{a^2-x^2}} y\,(a^2-x^2-y^2) \, dy$$

$$= \frac{1}{2} \int_0^a x \, dx \left[(a^2-x^2)\frac{y^2}{2} - y^4/4 \right]_0^{\sqrt{a^2-x^2}}$$

$$= \frac{1}{2} \int_0^a x \, dx \left[\frac{(a^2-x^2)(a^2-x^2)}{2} - \frac{(a^2-x^2)^2}{4} \right]$$

$$= \frac{1}{8} \int_0^a x\,(a^2-x^2)^2 \, dx.$$

If we let $u = a^2 - x^2$
$du = -2x\,dx \Rightarrow x\,dx = -\tfrac{1}{2}du$

and the integral becomes $\int -\frac{1}{2} u^2\,du = -\frac{u^3}{6}$

$$= -\frac{(a^2-x^2)^3}{6}.$$

Hence

$$= \frac{1}{8}\left[-\frac{(a^2-x^2)^3}{6}\right]_0^a$$

$$= +\frac{1}{48}\left[0 + a^6\right] = +\frac{a^6}{48}.$$

(b) The same integral in spherical co-ords where

$x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\theta$

$dV = r^2\sin\theta\,dr\,d\theta\,d\phi$

and the range of integration is

$0 \le r \le a$, $0 \le \theta \le \pi/2$ (hemisphere), $0 \le \phi \le \pi/2$ (first quadrant only).

$$\int_V xyz\,dV = \int_0^{\pi/2}\int_0^{\pi/2}\int_0^a r\sin\theta\cos\phi\, r\sin\theta\sin\phi\, r\cos\theta\, r^2\sin\theta\,dr\,d\theta\,d\phi$$

$$= \int_0^{\pi/2}\sin\phi\cos\phi\,d\phi \int_0^{\pi/2}\sin^3\theta\cos\theta\,d\theta \int_0^a r^5\,dr$$

$$= \underbrace{\int_0^{\pi/2}\sin\phi\cos\phi\,d\phi}_{}\ \underbrace{\int_0^{\pi/2}\sin^3\theta\cos\theta\,d\theta}_{}\ \left[\frac{r^6}{6}\right]_0^a$$

$u = \sin\phi$
$du = \cos\phi\,d\phi$
Integral becomes
$\int u\,du = \frac{u^2}{2}$

$u = \sin\theta$
$du = \cos\theta\,d\theta$
Integral becomes
$\int u^3\,du = u^4/4$

$$= \left[\frac{\sin^2\phi}{2}\right]_0^{\pi/2} \times \left[\frac{\sin^4\theta}{4}\right]_0^{\pi/2} \times \left[\frac{a^6}{6}\right]$$

$$= \left[\frac{1}{2} - 0\right] \times \left[\frac{1}{4} - 0\right] \times \frac{a^6}{6}$$

$$= \frac{a^6}{48}$$ which is the same answer as before!

Q3.

$$\underline{A} = x^2yz\,\underline{i} + xy^2z\,\underline{j} + xyz^2\,\underline{k}.$$

(a) $\oint_S \underline{A}\cdot d\underline{S}$

From question 1 the surface is a part of a sphere of radius a located in the first octant and bounded by the yz-plane $(x=0)$, the xz-plane $(y=0)$ and the xy-plane $(z=0)$.

Note that the vector field vanishes for $x=0$, $y=0$ and $z=0$. Hence the only nonzero contribution to this surface integral $\oint_S \underline{A}\cdot d\underline{S}$ comes from the spherical part of the surface where $d\underline{S} = \underline{e}_r\, dS$ and

$$\underline{e}_r = \sin\theta\cos\phi\,\underline{i} + \sin\theta\sin\phi\,\underline{j} + \cos\theta\,\underline{k}.$$

is the unit vector in the radial direction and

$$dS = a^2\sin\theta\, d\theta\, d\phi$$

is the surface area element in spherical co-ordinates for a sphere of radius a. Using $x = a\sin\theta\cos\phi$, $y = a\sin\theta\sin\phi$, $z = a\cos\theta$ for a point on the sphere of radius a, we find

$$\underline{A}\cdot\underline{e}_r = x^2yz\sin\theta\cos\phi + xy^2z\sin\theta\sin\phi + xyz^2\cos\theta$$

$$= a^4\sin^4\theta\cos\theta\cos^3\phi\sin\phi + a^4\sin^4\theta\cos\theta\sin^3\phi\cos\phi + a^4\sin^2\theta\cos^3\theta\sin\phi\cos\phi.$$

and $\oint_S \underline{A}\cdot d\underline{S} = \oint_S \underline{A}\cdot\underline{e}_r\, a^2\sin\theta\, d\theta\, d\phi$

$$= a^6 \int_0^{\pi/2}\int_0^{\pi/2} \left(\sin^5\theta\cos\theta\cos^3\phi\sin\phi + \sin^5\theta\cos\theta\sin^3\phi\cos\phi + \sin^3\theta\cos^3\theta\sin\phi\cos\phi\right) d\theta d\phi.$$

The integrals involved are

$$\underbrace{\int_0^{\pi/2} \sin^5\theta\cos\theta\, d\theta}_{\substack{u=\sin\theta \\ du=\cos\theta\, d\theta \\ \int u^5 du = \frac{u^6}{6}}} \underbrace{\int_0^{\pi/2} \cos^3\phi\sin\phi\, d\phi}_{\substack{u=\cos\phi \\ du=-\sin\phi\, d\phi \\ -\int u^3 du = -\frac{u^4}{4}}}$$

$$= \left[\frac{\sin^6\theta}{6}\right]_0^{\pi/2} \times \left[-\frac{\cos^4\phi}{4}\right]_0^{\pi/2} = \frac{1}{6}\times\frac{1}{4} = \frac{1}{24}.$$

and

$$\int_0^{\pi/2} \sin^5\theta\cos\theta\, d\theta \int_0^{\pi/2} \sin^3\phi\cos\phi\, d\phi$$

$$= \left[\frac{\sin^6\theta}{6}\right]_0^{\pi/2} \times \left[\frac{\sin^4\phi}{4}\right]_0^{\pi/2} = \frac{1}{6}\times\frac{1}{4} = \frac{1}{24}$$

and

$$\int_0^{\pi/2} \sin^3\theta\cos^3\theta\, d\theta \int_0^{\pi/2} \sin\phi\cos\phi\, d\phi$$

$$= \int_0^{\pi/2} \sin^3\theta(1-\sin^2\theta)\cos\theta\, d\theta \int_0^{\pi/2} \sin\phi\cos\phi\, d\phi$$

$$= \left[\int_0^{\pi/2} \sin^3\theta\cos\theta\, d\theta - \int \sin^5\theta\cos\theta\, d\theta\right] \int_0^{\pi/2} \sin\phi\cos\phi\, d\phi$$

$$\left[\frac{\sin^4\theta}{4} - \frac{\sin^6\theta}{6}\right]_0^{\pi/2} \times \left[\frac{\sin^2\phi}{2}\right]_0^{\pi/2}$$

$$= \left[\frac{1}{4} - \frac{1}{6}\right] \times \left[1/2\right] = \frac{2}{24}\times\frac{1}{2} = \frac{1}{24}$$

Hence the integral in question is

$$\oint_S \underline{A}\cdot d\underline{S} = a^6\left[1/24 + 1/24 + 1/24\right]$$

$$= \frac{3}{24}a^6 = \frac{a^6}{8}.$$

(b)
$$\nabla . \underline{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$
$$= \frac{\partial}{\partial x}(x^2 y z) + \frac{\partial}{\partial y}(x y^2 z) + \frac{\partial}{\partial z}(x y z^2)$$
$$= 2xyz + 2xyz + 2xyz$$
$$= 6xyz.$$

By Gauss's theorem $\oint_S \underline{A} . d\underline{S} = \int_V \nabla . \underline{A} \, dV$

where V is the volume enclosed by S. In our case the right-hand side is

$$\int_V \nabla . \underline{A} \, dV = 6 \int_V xyz \, dV$$

using the result from Question 2

$$\int_V \nabla . \underline{A} \, dV = 6 \left[\frac{a^6}{48} \right] = \frac{a^6}{8}$$

Hence $\oint_S \underline{A} . d\underline{S} = \frac{a^6}{8}$ which is the same answer that we found for part (a).

Q4.

(a). $\underline{B} = y^2 \underline{i} + z^2 \underline{j} + x^2 \underline{k}$.

i hemisphere $x^2+y^2+z^2=a^2$, $z \geq 0$.

On the surface of the hemisphere

$$\underline{B} \cdot d\underline{S} = \underline{B} \cdot \underline{e}_r dS$$

where $\underline{e}_r$ is given by

$$\underline{e}_r = \sin\theta\cos\phi\,\underline{i} + \sin\theta\sin\phi\,\underline{j} + \cos\theta\,\underline{k}.$$

and
$dS = a^2 \sin\theta\, d\theta\, d\phi$ where a is the radius of the sphere

and
$x = a\sin\theta\cos\phi$, $y = a\sin\theta\sin\phi$, $z = a\cos\theta$ are the co-ordinates on the surface of the sphere

Hence

$$\underline{B} \cdot \underline{e}_r = (y^2\underline{i} + z^2\underline{j} + x^2\underline{k}) \cdot (\sin\theta\cos\phi\,\underline{i} + \sin\theta\sin\phi\,\underline{j} + \cos\theta\,\underline{k})$$

$$= y^2 \sin\theta\cos\phi + z^2 \sin\theta\sin\phi + x^2\cos\theta$$

$$= a^2\sin^3\theta\sin^2\phi\cos\phi + a^2\cos^2\theta\sin\theta\sin\phi + a^2\sin^2\theta\cos\theta\cos^2\phi.$$

$$\therefore \int_S \underline{B} \cdot d\underline{S} = \int_0^{2\pi}\int_0^{\pi/2} \underline{B} \cdot \underline{e}_r dS$$

$$= \int_0^{2\pi}\int_0^{\pi/2} (a^2\sin^3\theta\sin^2\phi\cos\phi + a^2\cos^2\theta\sin\theta\sin\phi + a^2\sin^2\theta\cos\theta\cos^2\phi) * a^2\sin\theta\, d\theta\, d\phi$$

$$= a^4 \left[\int_0^{2\pi} \sin^2\phi\cos\phi\, d\phi \int_0^{\pi/2} \sin^4\theta\, d\theta + \int_0^{2\pi} \sin\phi\, d\phi \int_0^{\pi/2} \cos^2\theta\sin\theta\, d\theta + \int_0^{2\pi}\cos^2\phi\, d\phi \int_0^{\pi/2} \sin^3\theta\cos\theta\, d\theta \right]$$

In the last expression, the first two terms in the square brackets vanish, since

$$\int_0^{2\pi} \sin^2\phi \cos\phi \, d\phi = \left[\frac{\sin^3\phi}{3}\right]_0^{2\pi} = 0$$

and

$$\int_0^{2\pi} \sin\phi \, d\phi = \left[-\cos\phi\right]_0^{2\pi} = 0.$$

Hence

$$\int_S \underline{B}.d\underline{S} = a^4\left[\int_0^{2\pi}\cos^2\phi \, d\phi \int_0^{\pi/2} \sin^3\theta \cos\theta \, d\theta\right].$$

$$\int_0^{2\pi} \cos^2\phi \, d\phi = \frac{1}{2}\int_0^{2\pi}(1+\cos 2\phi)\, d\phi$$

$$= \frac{1}{2}\left[\phi + \frac{\sin 2\phi}{2}\right]_0^{2\pi} = \pi$$

and

$$\int_0^{\pi/2} \sin^3\theta \cos\theta \, d\theta = \int_0^{\pi/2} \sin^3\theta \, d(\sin\theta)$$

$$= \left[\frac{\sin^4\theta}{4}\right]_0^{\pi/2} = 1/4$$

Hence

$$\int_S \underline{B}.d\underline{S} = \frac{\pi a^4}{4}.$$

ii For the disk in the $z=0$ plane, $d\underline{S}$ is along the z-axis

$$\therefore d\underline{S} = \underline{k}\, dS$$

So we have

$$\int_S \underline{B}.d\underline{S} = \int_S \underline{B}.\underline{k}\, dS = \int_S x^2 \, dS$$

To integrate over the disk, we use 2D plane polar co-ordinates

$$x = r\cos\theta, \quad y = r\sin\theta$$

$$dS = r\,dr\,d\theta$$

$$0 \le r \le a, \quad 0 \le \theta \le 2\pi \quad \text{so that}$$

$$\int_S \underline{B}\cdot d\underline{s} = \int_0^{2\pi}\int_0^a r^2\cos^2\theta\, r\,dr\,d\theta$$

$$= \int_0^{2\pi}\cos^2\theta\, d\theta \int_0^a r^3\,dr$$

$$= \frac{1}{2}\int_0^{2\pi}(1+\cos 2\theta)\,d\theta \left[r^4/4\right]_0^a$$

$$= \frac{1}{2}\left[\theta + \frac{1}{2}\sin 2\theta\right]_0^{2\pi} a^4/4$$

$$= \frac{1}{2}\left[2\pi + 0 - 0\right] a^4/4 = \pi a^4/4$$

as before in i.

(b) $\underline{A} = \frac{1}{3}(z^3\underline{i} + x^3\underline{j} + y^3\underline{k})$

$c:\quad x^2+y^2=a^2, \quad z=0.$

To evaluate the integral $\oint_C \underline{A}\cdot d\underline{r}$ over the circle $x^2+y^2=a^2$, $z=0$, we parameterise it

$x = a\cos t, \quad y = a\sin t, \qquad 0 \le t \le 2\pi.$

$dx = -a\sin t\,dt, \quad dy = a\cos t\,dt.$

and $d\underline{r} = dx\,\underline{i} + dy\,\underline{j} + dz\,\underline{k}$

$= -a\sin t\,dt\,\underline{i} + a\cos t\,dt\,\underline{j} + 0\,\underline{k}.$

with $z=0$

$\underline{A} = \frac{1}{3}(x^3\underline{j} + y^3\underline{k})$

$\therefore \underline{A}\cdot d\underline{r} = \frac{1}{3}x^3 a\cos t\,dt = \frac{1}{3}a^4\cos^4 t.$

Hence

$$\oint_C \underline{A}\cdot d\underline{r} = \frac{a^4}{3}\int_0^{2\pi}\cos^4 t\,dt$$

$$= \frac{a^4}{3}\int_0^{2\pi}\left[\frac{1}{2}(1+\cos 2t)\right]^2 dt$$

$$= \frac{a^4}{3}\int_0^{2\pi}\frac{1}{4}(1+2\cos 2t+\cos^2 2t)\,dt$$

$$= \frac{a^4}{12}\left[\int_0^{2\pi}dt + 2\int_0^{2\pi}\cos 2t\,dt + \frac{1}{2}\int_0^{2\pi}(1+\cos 4t)\,dt\right]$$

$$= \frac{a^4}{12}\left\{ 2\pi + \left[\sin 2t\right]_0^{2\pi} + \frac{1}{2}\left[t + \tfrac{1}{4}\sin 4t\right]_0^{2\pi}\right\}$$

$$= \frac{a^4}{12}\left\{ 2\pi + 0 + \frac{1}{2}\left[2\pi + 0 - 0\right]\right\}$$

$$= \frac{a^4}{12}\left\{3\pi\right\} = \frac{\pi a^4}{4}$$ as before in i. and ii.

(c) $\underline{A} = \frac{1}{3}\left(z^3 \underline{i} + x^3 \underline{j} + y^3 \underline{k}\right)$

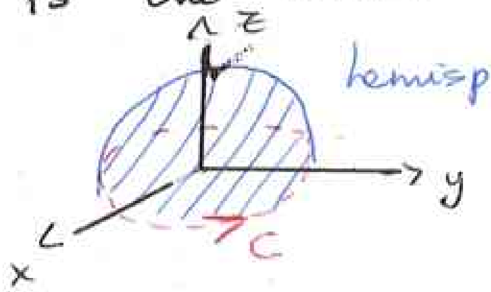
$$\nabla \times \underline{A} = \frac{1}{3}\begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ z^3 & x^3 & y^3 \end{vmatrix}$$

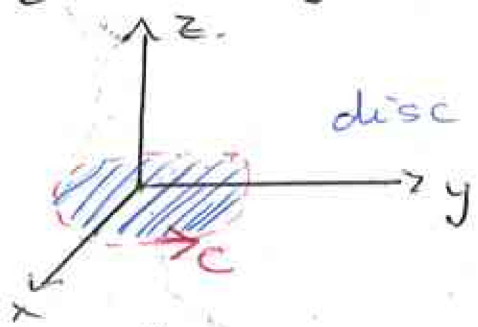
$$= \frac{1}{3}\left[\underline{i}\left(3y^2 - 0\right) - \underline{j}\left(0 - 3z^2\right) + \underline{k}\left(3x^2 - 0\right)\right]$$

$$= \frac{1}{3}\left[3y^2\underline{i} + 3z^2\underline{j} + 3x^2\underline{k}\right]$$

$$= y^2\underline{i} + z^2\underline{j} + x^2\underline{k}.$$

Hence $\nabla \times \underline{A} = \underline{B}$ from part (a). For both surfaces considered in (a), the circle $x^2+y^2=a^2$, $z=0$ is the boundary C (see diagram)





By Stoke's theorem $\oint_C \underline{A}\cdot d\underline{r} = \int_S \nabla\times\underline{A}\cdot d\underline{S}$

for any surface S bounded by the curve C. Hence the integral $\int_S \nabla\times\underline{A}\cdot d\underline{S} = \int_S \underline{B}\cdot d\underline{S}$ over any such surface, has the same value $\pi a^4/4$ equal to the line integral $\oint_C \underline{A}\cdot d\underline{r}$.